

SKYLAR PIETRASZEWSKI

JUNIOR BACKEND / SOFTWARE ENGINEER | PYTHON - APIS - DATA SYSTEMS - TESTING

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SUMMARY

Early-career software engineer with strong systems reasoning and experience developing Nova, an independent quantitative research and paper-execution platform. Skilled in requirements decomposition, API and data-pipeline design, debugging across component boundaries, test strategy, provenance, and fail-closed validation.

TECHNICAL SKILLS

Core development: Python, SQL, FastAPI, REST APIs, SQLite

Systems and data: Data pipelines, structured processing, provenance, validation, failure-state design

Testing and engineering: Git, automated/unit/integration testing, debugging, API design, failure-mode analysis

Application stack: React-based interfaces, backend/frontend integration

Additional: C# coursework, AutoCAD, 3-axis CNC programming and operation

PROJECT

Nova - Independent Software Engineering Project | 2024-Present

- Architected and iteratively developed a Python/FastAPI/SQLite/React quantitative research and paper-execution system organized around explicit provenance, controlled handoffs, validation, and human authority.
- Built a seven-stage governed data pipeline spanning ingestion, cleaning, enrichment, features, regime analysis, research output, and decision artifacts.
- Grew and maintained an audited suite of approximately 2,900 passing tests; added integration coverage after multi-stage runtime failures escaped unit tests.
- Traced systemic defects including swallowed exceptions producing false zero values, UTC/Eastern time mismatches, disconnected configuration, duplicate classifiers, and post-fill failures registering as success.
- Integrated real Interactive Brokers data and paper execution; audited one observation through a 14-link evidence chain from ingestion to a future learning signal.
- Audited scope: approximately 29,000-35,000 lines of Python across approximately 200 modules, 25 database tables, and 156 REST routes. No real capital was deployed and no strategy was cleared for it.

EXPERIENCE

Robert Scott Designs - CNC Programmer | February 2022-August 2024

Translated blueprints and specifications into manufacturable AutoCAD designs; prepared and operated 3-axis CNC equipment for furniture, cabinetry, architectural components, and signage.

Thoren Restorations - Chimney Technician Lead | 2019-2022

Diagnosed field issues, scoped work, completed repairs and installations, communicated findings to customers, and trained or led technicians.

EDUCATION & CERTIFICATION

Western Governors University - Software Engineering coursework, May 2023-May 2025

Google IT Support Professional Certificate - Issued March 2024